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Bidirectional data transmission over isolation medium

Abstract

An apparatus includes a controller having differential modulation control outputs and is configured to provide differential modulation signals having a particular frequency at the differential modulation control outputs. A differential modulator circuit is coupled between first differential terminals and second differential terminals. The differential modulator circuit has differential modulation control inputs coupled to the differential modulation control outputs. The differential modulator circuit is configured to: modulate first differential signals at the first differential terminals with the differential modulation signals having the particular frequency, provide the modulated first differential signals at the second differential terminals, modulate second differential signals at the second differential terminals with the differential modulation signals having the particular frequency, and provide the modulated second differential signals at the first differential terminals.

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Background/Summary

BACKGROUND

(1) In some applications, two devices with different voltage domains may need to communicate with each other via various medium and protocols. For example, two devices may be connected via a cable, a pair of connectors, or other types of medium, and communicate by transmitting data signals according to a protocol, such as Universal Serial Bus (USB) protocol, BASE100/1000-T1 Ethernet protocols, Data Over Cable Service Interface Specification (DOCSIS), etc., over the medium. For some applications, it may be desirable to interpose an isolation device between the two devices to provide galvanic isolation. Galvanic isolation can prevent a flow of direct current (DC) signal or low frequency signal between the two devices, which can mitigate potential circuit damage and safety hazard caused by such signals. However, the isolation medium may complicate the ability of the systems in performing bidirectional data transmission over the isolation medium and processing of the received data.

SUMMARY

(2) In one example, an apparatus includes a controller having differential modulation control outputs and is configured to provide differential modulation signals having a particular frequency at the differential modulation control outputs. A differential modulator circuit is coupled between first differential terminals and second differential terminals. The differential modulator circuit has differential modulation control inputs coupled to the differential modulation control outputs. The

differential modulator circuit is configured to: modulate first differential signals at the first differential terminals with the differential modulation signals having the particular frequency, provide the modulated first differential signals at the second differential terminals, modulate second differential signals at the second differential terminals with the differential modulation signals having the particular frequency, and provide the modulated second differential signals at the first differential terminals.

(3) In another example, an apparatus includes a first controller having first differential modulation control outputs and is configured to provide first differential modulation signals having a particular frequency at the first differential modulation control outputs. A first differential modulator circuit is coupled between first differential terminals and second differential terminals. The first differential modulator circuit has first differential modulation control inputs coupled to the first differential modulation control outputs. The first differential modulator circuit is configured to: modulate first differential signals at the first differential terminals with the first differential modulation signals having the particular frequency, provide the modulated first differential signals at the second differential terminals, modulate modulated second differential signals at the second differential terminals with the first differential modulation signals having the particular frequency to recover second differential signals, and provide the recovered second differential signals at the first differential terminals. A second controller has second differential modulation control outputs and is configured to provide second differential modulation signals having the particular frequency at the second differential modulation control outputs. A second differential modulator circuit is coupled between third differential terminals and fourth differential terminals. The second differential modulator circuit has second differential modulation control inputs coupled to the second differential modulation control outputs. The second differential modulator circuit is configured to modulate the modulated first differential signals at the third differential terminals with the second differential modulation signals having the particular frequency to recover the first differential signals, provide the recovered first differential signals at the fourth differential terminals, modulate second differential signals at the fourth differential terminals with the second differential modulation signals having the particular frequency, and provide the modulated second differential signals at the third differential terminals.

(4) In yet another example, an apparatus includes a controller having a modulation control output and is configured to provide a modulation signal having a particular frequency at the modulation control output. A modulator circuit is coupled between a first terminal and a second terminal. The modulator circuit has a modulation control input coupled to the modulation control output. The modulator circuit is configured to: modulate a first signal at the first terminal with the modulation signal having the particular frequency; provide the modulated first signal at the second terminal; modulate a second signal at the second terminal with the modulation signal having the particular frequency, and provide the modulated second signal at the first terminal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of a system that supports bidirectional data communication over an isolation device, in accordance with various examples.

(2) FIG. 2 is a schematic diagram of internal components of FIG. 1 to support bidirectional data communication over an isolation medium, in accordance with various examples.

(3) FIG. 3 are graphs illustrating operations of the internal components of FIG. 1 in frequency domain, in accordance with various examples.

(4) FIG. 4 are graphs illustrating operations of the internal components of FIG. 1 in time domain, in accordance with various examples.

- (5) FIGS. 5, 6, 7, 8, and 9 are schematic diagrams of internal components of FIG. 1 to support bidirectional data communication over an isolation medium, in accordance with various examples.
- (6) FIGS. 10, 11, 12, and 13 are schematic diagrams of an isolation device that can be part of the system of FIG. 1, in accordance with various examples.
- (7) FIG. 14 is a schematic diagram of internal components of FIG. 1 to support bidirectional data communication over an isolation medium, in accordance with various examples.

DETAILED DESCRIPTION

- (8) The same reference numbers or other reference designators are used in the drawings to designate the same or similar (either by function and/or structure) features.
- (9) FIG. 1 is a schematic diagram of a system **100** that supports bidirectional data communication over an isolation device, in accordance with various examples. System **100** includes a first device **110** and a second device **130**, and an isolation device **150** coupled between the first device **110** and the second device **130**. Isolation device **150** can provide galvanic isolation between the first device **110** and the second device **130**. The first device **110** can include a controller **112** and a transceiver **114**. First device **110** can operate in a first power/voltage domain, for which first device **110** has a power supply terminal **115** that receives a supply voltage **PW1**, and a supply reference terminal **116** coupled to a first ground **GND1**. Also, second device **130** can include a controller **132** and a transceiver **134**. Second device **130** can operate in a second power/voltage domain, for which second device **130** has a power supply terminal **135** that receives a supply voltage **PW2**, and a supply reference terminal **136** coupled to a second ground **GND2**. The supply voltage **PW1** with respect its ground **GND1** can be the same or different voltage than the supply voltage **PW2** with respect to its ground **GND2**. In some examples, supply voltage **PW1** can be 5-20V DC and supply voltage **PW2** can be 1 kV DC.
- (10) Isolation device **150** can provide galvanic isolation between power supply terminals **115** and **135**, and between first ground **GND1** and second ground **GND2**. Isolation device **150** can have a band pass characteristic, and can prevent (or at least reduce) the flow of DC/low frequency current and/or voltage signals outside the pass frequency band of isolation device **150**. Isolation device **150** can include a transformer and can prevent flow of DC/low frequency current and/or voltage signals between power supply terminals **115** and **135**, and between first ground **GND1** and second ground **GND2**. Such arrangements can prevent or mitigate potential circuit damage and safety hazard caused by such signals. For example, first device **110** may be a consumer electronic device handled by a person and can tolerate only a low supply voltage **PW1** (e.g., 5-20V DC), and second device **110** may be an industrial sensor operating with a high supply voltage **PW2** (e.g., 1 kV DC). Isolation device **150** can prevent a large voltage signal and/or a large current signal from propagating from supply reference terminal **135** to supply reference terminal **115**, and/or from supply reference terminal **136** to supply reference terminal **116**, which may otherwise damage first device **110** and/or pose safety hazard to the person handling first device **110**.
- (11) Also, transceiver **114** of the first device **110** can include differential terminals **225** and **226** on which transceiver **114** can transmit or receive differential signals **D+/D-**, respectively. Differential terminals **225** and **226** can be coupled to corresponding differential terminals **151** and **152** of isolation device **150**. Transceiver **134** of the second device **130** can include differential terminals **265** and **266** on which transceiver **134** can transmit or receive differential signals **D+/D-** respectively. Differential terminals **265** and **266** can be coupled to corresponding differential terminals **153** and **153** of isolation device **150**. Each of transceiver **114** and **134** can include a differential receiver that extract the information represented by the differential signals **D+/D-** by subtracting between **D+** and **D-**.
- (12) The differential signals are alternating current (AC) signals having a relatively high frequency component and can be within the pass frequency band of isolation device **150**, so that isolation device **150** can transmit the differential signals between first device **110** and second device **130** with no or reduced attenuation. On the other hand, isolation device **150** can prevent the

transmission of low frequency component of the differential signals, such as the common mode or bias voltage of the differential signals. But because the differential receiver in each of transceiver **114** and **134** extract the information represented by the differential signals D+/D- by subtracting between D+ and D-, the common mode component of D+ and D- is largely absent in the extracted information, and the removal of the DC or low frequency common mode component from D+ and D- by isolation device **150** does not affect the information extraction.

(13) In some examples, to facilitate transmission of the differential signals via isolation device **150**, each of devices **110** and **130** can perform a modulation operation on the differential signals, and transmit the modulated differential signals via isolation device. For example, first device **110** can modulate first differential signals to be transmitted using a first modulation signal having a particular frequency, and transmit the modulated first differential signals to differential terminals **225** and **256** (and differential terminals **151** and **152**). Isolation device **150** can receive the modulated first differential signals at differential terminals **151** and **152**, and transmit the modulated first differential signals to differential terminals **153** and **154** (and differential terminals **265** and **266**). Second device **130** can receive the modulated first differential signals at differential terminals **265** and **266**, and modulate the modulated first differential signals using a second modulation signal (as part of a demodulation operation) having the particular frequency to recover the first differential signals.

(14) Also, second device **130** can modulate second differential signals to be transmitted using the second modulation signal, and transmit the modulated second differential signals to differential terminals **265** and **256** (and differential terminals **153** and **154**). Isolation device **150** can receive the modulated second differential signals at differential terminals **153** and **154**, and transmit the modulated second differential signals to differential terminals **151** and **152** (and differential terminals **225** and **226**). First device **110** can receive the modulated second differential signals at differential terminals **225** and **226**, and modulate the modulated second differential signals using the first modulation signal (as part of a demodulation operation) to recover the second differential signals.

(15) The modulation operation can shift the first/second differential signals from a first frequency band to a second frequency band, where the second frequency band has a reduced ratio between the maximum and minimum frequencies compared with the first frequency band. Isolation device **150** can have a pass frequency band that matches or includes the second frequency band. Such arrangements can relax the transformer design of isolation device **150** by reducing the maximum and minimum frequencies of the pass frequency band of isolation device **150**, and allow transmission of the differential signals with no or reduced attenuation via isolation device **150** having such pass band characteristics. Specifically, a transformer designed for higher frequency operation can require smaller primary/secondary coil inductance values, which can be achieved with a smaller number of turns of each coil. A transformer with fewer turns of the primary and second coils is smaller than a transformer with more turns, all else being equal. Further, a transformer with fewer turns of the primary and secondary coils can be implemented with a higher quality factor and at a lower cost.

(16) Also, the modulation and demodulation operation can be largely agnostic to the signaling protocol and bidirectional transmission mode of the differential signals, so long as the frequency band of the modulated signal is within the pass frequency band of isolation device **150**. Accordingly, differential signals of various protocols (e.g., USB, BASE100/1000-T1 Ethernet, DOCSIC, etc.) and of various bidirectional transmission modes (e.g., full duplex and half duplex) can be modulated for transmission over isolation device **150** and demodulated to recover the differential signals.

(17) FIG. 2 is a schematic diagram illustrating internal components of system **100** in accordance with various examples. In the example illustrated in FIG. 2, transceiver **114** can include a data source/sink **202**, a low pass filter (LPF) **204**, and a modulation circuit **206**. In another example,

data source/sink **202** is external to the transceiver **114**. Low pass filter **204** is coupled between data source/sink **202** and modulation circuit **206**. Low pass filter **204** has differential filter terminals **221** and **222**. Modulation circuit **206** can include transistors **211**, **212**, **213**, and **214**. In some examples, transistors **211-214** can each include High Electron Mobility Transistors (HEMTs). Differential filter terminal **221** of low pass filter **204** can be coupled to first current terminals (e.g., sources) of transistors **211** and **212**. Differential filter terminal **222** can be coupled to first current terminals (e.g., sources) of transistors **213** and **214**. Second current terminals (e.g., drains) of transistors **211** and **213** can be coupled to the differential terminal **225**, and second current terminals (e.g., drains) of transistors **212** and **214** can be coupled to the differential terminal **226**.

(18) Also, controller **112** has control inputs **209** and **210** and differential modulation control outputs **201** and **203**. Differential modulation control output **201** can be coupled to the gates of transistors **211** and **214**, and differential modulation control output **203** can be coupled to the gates of transistors **212** and **213**. Controller **112** generates differential modulation signals CLKP1 and CLKN1 at differential modulation control outputs **201** and **203** as shown. The differential modulation signals CLKP1/CLKN1 can have different polarities and each has a frequency $f_{\text{sub.m}}$ and a cycle period $T_{\text{sub.m}}$ ($T_{\text{sub.m}}$ is $1/f_{\text{sub.m}}$) programmed according to a signal at control input **210**. CLKN1 is 180 degrees out of phase with respect to CLKP1. When CLKP1 is logic high and CLKN1 is logic low, transistors **211** and **214** are on and transistors **212** and **213** are off, and when CLKP1 is logic low and CLKN1 is logic high, transistors **211** and **214** are off and transistors **212** and **213** are on.

(19) Also, transceiver **134** can include a data source/sink **232**, an LPF **234**, and a modulation circuit **236**. In another example, data source/sink **232** is external to the transceiver **134**. Low pass filter **234** is coupled between data source/sink **232** and modulation circuit **236**. Low pass filter **234** has differential filter terminals **261** and **262**. Modulation circuit **236** can include transistors **241**, **242**, **243**, and **244**. Transistors **241-244** can be FETs/HEMTs. Differential filter terminal **261** can be coupled to the first current terminals (e.g., sources) of transistors **241** and **242**. Differential filter terminal **262** can be coupled to the first current terminals (e.g., sources) of transistors **243** and **244**. Second current terminals (e.g., drains) of transistors **241** and **243** can be coupled to the differential terminal **265**, and the second current terminals (e.g., drains) of transistors **242** and **244** can be coupled to the differential terminal **266**. Controller **132** has a control input **229**, a control input **230**, and differential modulation control outputs **251** and **252**. Differential modulation control output **251** can be coupled to the gates of transistors **241** and **244**, and differential modulation control output **252** can be coupled to the gates of transistors **242** and **243**. Controller **132** generates differential modulation signals CLKP2 and CLKN2 at its differential modulation control outputs **251** and **252** as shown. The differential modulation signals CLKP2/CLKN2 can have different polarities and each has the same frequency $f_{\text{sub.m}}$ and the same cycle period $T_{\text{sub.m}}$ ($T_{\text{sub.m}}$ is $1/f_{\text{sub.m}}$) as CLKP1/CLKN1 and can be programmed according to a signal at control input **230**. CLKN2 is 180 degrees out of phase with respect to CLKP2. When CLKP2 is logic high and CLKN2 is logic low, transistors **241** and **244** are on and transistors **242** and **243** are off, and when CLKP2 is logic low and CLKN2 is logic high, transistors **241** and **244** are off and transistors **242** and **243** are on.

(20) In one example operation, data source/sink **202** can generate differential signals $P1+/P1-$, which is provided to LPF **204**. LPF **204** filters differential signal $P1+/P1-$ and provides filtered differential signals $V1+$ and $V1-$ to the modulation circuit **206**. Modulation circuit **206** modulates the filtered differential signals $V1+/V1-$ using differential modulation signals CLKP1/CLKN1 to produce modulated differential signals $V2+/V2-$. The frequency of the differential modulation signals CLKP1/CLKN1 produced by controller **112** is higher than the frequency of the baseband filtered differential signal $V1+/V1-$. In one example, the frequency of the differential modulation signals CLKP1/CLKN1 is n -times higher than the frequency of the filtered differential signals $V1+/V1-$, where n can be 2, 3, 4, 5, 6, 7, 8, etc.

(21) Modulated differential signals $V2+/V2-$ are provided to differential terminals **151** and **152** of isolation device **150**. Isolation device **150** transmit the modulated differential signal $V2+/V2-$ from differential terminals **151/152** to differential terminals **153/154** as modulated differential signals $V4+/V4-$, and thus to differential terminals **265/266** of modulation circuit **236**. Modulation circuit **236** mixes the modulated differential signals $V4+/V4-$ with differential modulation signal $CLKP2/CLKN2$ from controller **132** to produce differential signals $V3+/V3-$ as part of a demodulation operation. LPF **234** low pass filters differential signals $V3+/V3-$ to recover differential signal $P2+/P2-$, which is provided to data source/sink **232**.

(22) Also, in one example operation, data source/sink **232** can generate differential signals $P2+/P2-$, which can be provided to LP **234**. LPF **234** filters differential signal $P2+/P2-$ and provides filtered differential signal $V3+/V3-$ to the modulation circuit **236**. Modulation circuit **236** modulates the differential signals $V3+/V3-$ using differential modulation signals $CLKP2/CLKN2$ to produce modulated differential signals $V4+/V4-$. The frequency of differential modulation signals $CLKP2/CLKN2$ is identical to differential modulation signals $CLKP1/CLKN1$ provided to modulation circuit **206** and is higher than the frequency of the filtered differential signals $V3+/V3-$. Modulated differential signals $V4+/V4-$ are provided to differential terminals **153** and **153** of isolation device **150**, which transmits the modulated differential signal $V4+/V4-$ from differential terminals **153/154** to differential terminals **151/152** as modulated differential signals $V2+/V2-$, and thus to differential terminals **225/226** of modulation circuit **206**. Modulation circuit **206** mixes the modulated differential signal $V2+/V2-$ with differential modulation signals $CLKP1/CLKN1$ from controller **132** as part of a demodulation operation to produce differential signal $V1+/V1-$. LPF **204** low pass filters differential signal $V1+/V1-$ to recover differential signals $P1+/P1-$, which are provided to data source/sink **202**.

(23) As described above, the frequency of the differential modulation signals $CLKP1/CLKN1$ and $CLKP2/CLKN2$ can be the same. However, to account for the propagation delay ($\tau_{sub.f}$) through isolation device **150**, one of the controllers **112**, **132** imposes a delay, τ , in its differential modulation signal with respect to the other controller's differential modulation signals. For relatively low loss operation through isolation device **150**, the delay value t , the propagation delay $\tau_{sub.f}$, and a function $m(t)$ representing the differential modulation signals ($CLKP1/CLKN1$ and $CLKP2/CLKN2$) can be related as follows:

$$(24) \quad \text{Abs}(m(t - \tau_{sub.f}) * m(t - t)) = 1 \quad (\text{Eq. 1}) \quad \text{Abs}(m(t - \tau_{sub.f}) * m(t)) = 1 \quad (\text{Eq. 2})$$

(25) In some examples, the delay value t , the propagation delay $\tau_{sub.f}$, and the cycle period of the differential modulation signals modulation signal $T_{sub.m}$ can be related as follows to satisfy the conditions of Equations 1 and 2 above:

$$(26) \quad \tau_{sub.f} = t + l\left(\frac{T_m}{2}\right) \quad (\text{Eq. 3}) \quad \tau_{sub.f} = k\left(\frac{T_m}{4}\right) \quad (\text{Eq. 4})$$

(27) In Equations 3 and 4, k can include any non-zero integer, and l can include an integer including zero. As an example, in a case where the frequency of the modulation signal equals 8 GHz, the propagation delay $\tau_{sub.f}$ of isolation device **150** can be 31.25 picoseconds (ps). Either of the controllers **112** and **132** can be programmed with the delay τ through the respective control input **209** or **229** (e.g., by programming a register internal to the controller).

(28) FIG. 3 includes graphs **302** and **314** that illustrate an example modulation operation in frequency domain. Frequency graph **302** illustrate example differentials signals $V1+/V1-$ ($V1$ in FIG. 3) in frequency domain. In this example, the frequency band of the differential signals $V1$ is 100 MHz to 2 GHz, where the ratio between the maximum frequency and the minimum frequency of the frequency band is 20.

(29) Frequency graph **314** illustrates example modulated differential signals $V2+/V2-$ ($V2$ in FIG. 3) provided by modulation circuit **206** by modulating differentials signals $V1+/V1-$ using differential modulation signals $CLKP1/CLKN1$ having a frequency of 8 GHz. Because of the modulation, the frequency band of modulated differential signals $V2+/V2-$ is 6 GHz and 10 GHz,

where a ratio between the maximum frequency and the minimum frequency of the frequency band is 1.67. As explained above, the reduced ratio can relax the transformer design of isolation device **150**.

(30) FIG. 4 include graphs **401**, **402**, **403**, **404**, and **405** that illustrate example operations of modulator circuit **206**, isolation device **150**, and modulator circuit **236** in time domain. Graph **401** illustrates an example of differential signals $V1+/V1-$ (labelled as $V1$ in FIG. 4) provided by data source/sink **202** and transmitted by transceiver **114**. Graph **402** illustrates an example of differential modulation signals $CLKN1/CLKP1$ (labelled as $m.sub.0(t)$ in FIG. 4). Graph **403** illustrates example differential signals $V4+/V4-$ (labelled as $V4$ in FIG. 4) that are delayed from $V1$ by propagation delay $\tau.sub.f$ of isolation device **150**. Graph **404** illustrates example differential modulation signals $CLKN2/CLKP2$ that are phase shifted/delayed by the delay value τ (labelled as $m.sub.0(t-\tau)$ in FIG. 4). Further, graph **405** illustrates example differential signals $P2+/P2-$ (labelled as $P2$ in FIG. 4) obtained by mixing $V4$ with $m.sub.0(t-\tau)$ as part of a demodulation operation, followed by low pass filtering by LPF **234**. One of the controllers (e.g., controller **132**) implements the delay τ for the modulation signal $m(t)$ of waveform **404** relative to the modulation signal $m(t)$ of waveform **402**.

(31) FIG. 5 is a schematic diagram of internal components of system **100** in accordance with various examples. As described above, either controller **112** or **132** can implement the delay τ (as programmed through the corresponding control input **209** or **229**) for its differential modulation signal $CLKP/CLKN$ relative to the other controller's differential modulation signal. Each controller **112** and **132** in the example of FIG. 5 can generate differential quadrature (in-phase (I) and quadrature (Q)) modulation signals for $CLKP/CLKN$. The quadrature modulation signals can include $m.sub.0(t)$, $m.sub.90(t)$, $m.sub.180(t)$, and $m.sub.270(t)$, where the subscripts 0, 90, 180, and 270 refer to the four clock phases 0 degrees, 90 degrees, 180, degrees, and 270 degrees.

(32) Controller **112** has an I/Q control input **521**, and controller **132** has an I/Q control input **561**. In one example, a logic high on an I/Q control input **521** or **561** causes the corresponding controller to select the I modulation signal phases ($m.sub.0(t)$ and $m.sub.180(t)$) as its differential modulation signal $CLKP/CLKN$, and a logic low on an I/Q control input **521** or **561** causes that controller to select the Q modulation signal phases ($m.sub.90(t)$ and $m.sub.270(t)$) as its differential modulation signal $CLKP/CLKN$. The I/Q control inputs **521/561** of controllers **112/132** can be programmed such that the I phase for the modulation signals of one of the controllers is selected and the Q phase for the other controller is selected.

(33) If the I/Q control input **521** for controller **112** is programmed for the I phase and the I/Q control input **561** for controller **132** is programmed for the Q phase, controller **112** can generate $CLKP1$ to have phase $m.sub.0(t)$ and $CLKN1$ to have phase $m.sub.180(t)$ (the I phase), and controller **132** can generate $CLKP2$ to have phase $m.sub.90(t-\tau)$ and $CLKN2$ to have phase $m.sub.270(t-\tau)$ (the Q phase). In this configuration, controller **132** implements the delay t .

(34) If the I/Q control input **521** for controller **112** is programmed for the Q phase and the I/Q control input **561** for controller **132** is programmed for the I phase, controller **132** can generate $CLKP2$ to have phase $m.sub.0(t)$ and $CLKN2$ to have phase $m.sub.180(t)$ (the I phase), and controller **112** can generate $CLKP1$ to have phase $m.sub.90(t-\tau)$ and $CLKN1$ to have phase $m.sub.270(t-\tau)$ (the Q phase). In this configuration, controller **112** implements the delay t .

(35) FIG. 6 is a schematic diagram of system **100** that is mostly the same as system **100** in FIG. 2. Controllers **112** and **132** in the example of FIG. 6 have a delay enable control input-delay enable control input **621** for controller **112** and delay enable control input **661** for controller **132**. Each controller **112**, **132** can be programmed to enable delay τ or not to enable delay τ . For example, a logic high on delay enable control input **621**, **661** can program that controller to implement delay τ , and a logic low on the delay enable control can program the controller not implement delay τ . One of the controllers is programmed to implement delay τ , while the other controller is programmed not to implement delay τ .

(36) FIG. 6 also illustrates that controllers **112** and **132** implement clock injection. The oscillation frequencies of controllers **112** and **132** can be at similar but not sufficiently close frequencies when they are uncoupled. A strong coupling through a galvanic isolation capacitor **645** allows controller **132** to capture the oscillation frequency of controller **112** and results in controller **132** to oscillate substantially at an identical frequency, allowing the controllers **112** and **132** to be synchronized.

(37) FIG. 7 is a schematic diagram of system **100** similar to that shown in FIG. 2 but with each transceiver having a clock recovery circuit. Transceiver **114** includes a clock recovery circuit **810**, and transceiver **134** includes a clock recovery circuit **860**. Clock recovery circuit **810** includes inputs **811** and **812** coupled to differential filter terminals **221** and **222**, respectively. An output **813** of clock recovery circuit **810** is coupled to control input **209** of controller **112**. Clock recovery circuit **810** recovers the frequency and phase of controller **112** to synchronize the clock of controller **132** to controller **112**. A clock recovery loop can be implemented with a phase-locked-loop using non-data-aided or decision-directed carrier recovery techniques used in wireless communication systems. In addition to traditional carrier recovery approaches, a leaky carrier recovery as in U.S. Pat. No. 11,405,042, incorporated herein by reference, could be used. Similarly, clock recovery circuit **860** includes inputs **861** and **862** coupled to differential filter terminals **261** and **262**, respectively. An output **863** of clock recovery circuit **860** is coupled to control input **229** of controller **132**. Clock recovery circuit **880** also can recover a clock. An output **863** of clock recovery circuit **860** is coupled to control input **209** of controller **132**.

(38) FIG. 8 is a schematic diagram of controllers **112** and **132**. Each controller **112**, **132** can include an oscillator **902**, a differential signal generator **904**, and a programmable delay circuit **910**. An output **903** of oscillator **902** is coupled to an input **905** of the differential signal generator **904**. The differential signal generator **904** generates differential output signals at the respective outputs **0** and **180**, which are coupled to corresponding inputs **913**, **915** of the programmable delay circuit **910**. The programmable delay circuit **915** can generate the differential modulation signals CLKP/CLKN at its outputs **917** and **919**. For controller **112**, the outputs **917** and **919** of its programmable delay circuit **915** are (or are coupled to) the control outputs **201** and **203**. For controller **132**, the outputs **917** and **919** of its programmable delay circuit **915** are (or are coupled to) the control outputs **251** and **252**.

(39) The differential signal generator **904** receives an oscillation signal (e.g., a sinusoidal signal) from the oscillator **902** and converts the oscillation signal into differential square wave signals at its **0** and **180** outputs. The programmable delay circuit **910** can then add (or not add) the delay τ described above to generate the differential modulation signals CLKP/CLKN. The programmable delay circuit **910** is configured to implement a delay in accordance with the delay value provided at control input **209/229** or not to implement a delay if no delay value is provided at the control input.

(40) FIG. 9 is another schematic diagram of controllers **112** and **132**. Each controller **112**, **132** in this example can include oscillator **902**, an I/Q generator **1004**, the programmable delay circuit **910**, and multiplexers **1008** and **1010**. The output **903** of oscillator **902** is coupled to the input **905** of the differential signal generator **904**. The I/Q generator **1003** generates four quadrature differential output signals at the respective outputs **0**, **90**, **180**, and **270** outputs. Outputs **90** and **270** are coupled to inputs **913** and **915** of the programmable delay circuit **910**. Multiplexer **1008** has inputs **1001** and **1002**, a selection input **1005**, and an output **1003**. Multiplexer **1010** has inputs **1011** and **1012**, a selection input **1015**, and an output **1013**. Selection input **1015** is inverted with respect to selection input **1005**. The selection inputs **1005** and **1015** are coupled to control input **521/561**. The **0** and **180** outputs of the I/Q generator **1004** are coupled to inputs **1001** and **1011**, respectively, of multiplexers **1008** and **1010**. The outputs **917** and **919** of the programmable delay circuit **910** are coupled to inputs **1002** and **1012**, respectively, of multiplexers **1008** and **1010**. The programmable delay circuit **910** can introduce the aforementioned delay τ for the modulation signals $m_{\text{sub.90}}(t)$ and $m_{\text{sub.270}}(t)$, as described above.

(41) If the I/Q control input **521/561** is asserted to a first logic state (e.g., logic high), multiplexers

1008 and **1010** select their inputs **1001** and **1011** to produce the CLKP signal as m.sub.0(t) and the CLKN signal as m.sub.180(t). If the I/Q control input **521/561** is asserted to a second logic state (e.g., logic low), multiplexers **1008** and **1010** select their inputs **1001** and **1012** to produce the CLKP signal as m.sub.90(t) and the CLKN signal as m.sub.270(t), with or without delay τ based on the logic level of the control signal at control input **209/229**.

(42) FIGS. **10-12** are schematic diagrams illustrating various examples of isolation device **150**. Isolation device **150** in FIG. **10** includes capacitors **1101** and **1102** and a transformer **1103**. Each capacitor **1101**, **1102**, is coupled across a respective winding of transformer **1103**. Isolation device **150** in FIG. **10** can be a double tuned transformer.

(43) FIG. **11** includes capacitors **1201**, **1202**, and **1203** and transformers **1204** and **1206**. Capacitor **1201** is coupled across a winding of transformer **1204**, and capacitor **1202** is coupled across the opposing winding of transformer **1204** as well as across a winding of transformer **1206**. Capacitor **1203** is coupled across the opposing winding of transformer **1206**. Isolation device **150** in FIG. **11** can be a cascaded double tuned transformer.

(44) FIG. **12** includes capacitors **1301**, **1302**, **1303**, and **1304**, transformers **1305** and **1306**, and transmission lines **1307** and **1308**. Capacitor **1301** is coupled across a winding of transformer **1305**, and capacitor **1302** is coupled across the opposing winding of transformer **1305**. Capacitor **1303** is coupled across a winding of transformer **1306**, and capacitor **1304** is coupled across the opposing winding of transformer **1306**. Transmission line **1307** is coupled between the upper plates of capacitors **1302** and **1303**, and transmission line **1308** is coupled between the lower plates of capacitors **1302** and **1303**, as shown. Isolation device **150** in FIG. **12** can be a transmission line-coupled, double tuned transformer.

(45) The isolation device **150** of FIG. **13** is similar to that of FIG. **12** but capacitors **1405** and **1406** are used instead of transmission lines **1307** and **1308**. Isolation device **150** in FIG. **13** can be a capacitor-coupled, double tuned transformer.

(46) FIG. **14** is a schematic diagram illustrating internal components of system **100** in accordance with various examples. In the example of FIG. **14**, modulation circuit **206** can be a single-ended modulator circuit including transistor **211** controlled by modulation signal CLKP1, and modulation circuit **236** can be a single-ended modulator circuit including transistor **241** controlled by modulation signal CLKP2. Data source/sink **202** can provide or receive a single-ended signal P1+, and data source/sink **232** can provide or receive a single-ended signal P2+. Modulation circuit **206** can modulate the single-ended signal P1+ with modulation signal CLKP1, and modulation circuit **236** can modulate the single-ended signal P2+ with modulation signal CLKP2. Modulation signal CLKP1 can be provided by controller **112**, and modulation signal CLKP2 can be provided by controller **132**. The modulated signals can be transmitted over isolation device **150**. Modulation circuit **206** can also modulate the modulated signal P2+ received from isolation device **150** with CLKP1, and process the modulated signal P2+ with LPF **204** to recover P2+. Modulation circuit **236** can also modulate the modulated signal P1+ received from isolation device **150** with CLKP2, and process the modulated signal P1+ with LPF **234** to recover P1+. As described above, modulation signals CLKP1 and CLKP2 can have a same frequency but phase-shifted with respect to each other to account the propagation delay through isolation device **150**.

(47) In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

(48) Also, in this description, the recitation “based on” means “based at least in part on.” Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

(49) A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

(50) As used herein, the terms “terminal”, “node”, “interconnection”, “pin” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.

(51) A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

(52) While the use of particular transistors is described herein, other transistors (or equivalent devices) may be used instead with little or no change to the remaining circuitry. For example, a field effect transistor (“FET”) (such as an n-channel FET (NFET) or a p-channel FET (PFET)), a bipolar junction transistor (BJT—e.g., NPN transistor or PNP transistor), an insulated gate bipolar transistor (IGBT), and/or a junction field effect transistor (JFET) may be used in place of or in conjunction with the devices described herein. The transistors may be depletion mode devices, drain-extended devices, enhancement mode devices, natural transistors or other types of device structure transistors. Furthermore, the devices may be implemented in/over a silicon substrate (Si), a silicon carbide substrate (SiC), a gallium nitride substrate (GaN) or a gallium arsenide substrate (GaAs).

(53) References may be made in the claims to a transistor's control input and its current terminals. In the context of a FET, the control input is the gate, and the current terminals are the drain and source. In the context of a BJT, the control input is the base, and the current terminals are the collector and emitter.

(54) References herein to a FET being “ON” or “enabled” means that the conduction channel of the FET is present and drain current may flow through the FET. References herein to a FET being “OFF” or “disabled” means that the conduction channel is not present so drain current does not flow through the FET. An “OFF” FET, however, may have current flowing through the transistor's body-diode.

(55) Circuits described herein are reconfigurable to include additional or different components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the resistor shown. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.

(56) While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other example embodiments, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the

features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated. As used herein, the term “integrated circuit” means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

(57) Uses of the phrase “ground” in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.

(58) Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. An apparatus comprising: a controller having differential modulation control outputs and including a programmable delay circuit, the controller configurable to provide differential modulation signals having a particular frequency at the differential modulation control outputs; and a differential modulator circuit coupled between first differential terminals and second differential terminals, the differential modulator circuit having differential modulation control inputs coupled to the differential modulation control outputs, the differential modulator circuit configurable to: modulate first differential signals at the first differential terminals with the differential modulation signals having the particular frequency; provide the modulated first differential signals at the second differential terminals; modulate second differential signals at the second differential terminals with the differential modulation signals having the particular frequency; and provide the modulated second differential signals at the first differential terminals.

2. The apparatus of claim 1, wherein the differential modulator circuit includes: a first transistor coupled between a first one of the first differential terminals and a first one of the second differential terminals, the first transistor having a first control terminal coupled to a first one of the differential modulation control inputs; a second transistor coupled between the first one of the first differential terminals and a second one of the second differential terminals, the second transistor having a second control terminal coupled to a second one of the differential modulation control inputs; a third transistor coupled between a second one of the first differential terminals and the first one of the second differential terminals, the third transistor having a third control terminal coupled to the second one of the differential modulation control inputs; and a fourth transistor coupled between the second one of the first differential terminals and the second one of the second differential terminals, the fourth transistor having a fourth control terminal coupled to the first one of the differential modulation control inputs.

3. The apparatus of claim 2, wherein the controller has first and second control inputs and includes: an oscillator having a frequency control input and an oscillator output, the frequency control input coupled to the first control input, the oscillator configurable to set the particular frequency responsive to a state of the first control input; and a modulation signal generator having a generator input and differential generator outputs, the generator input coupled to the oscillator output; and wherein the programmable delay circuit is coupled between the differential generator outputs and the differential modulation control outputs, the programmable delay circuit has a delay control input coupled to the second control input.

4. The apparatus of claim 2, wherein the programmable delay circuit has differential delay inputs, differential delay outputs, and a delay control input, wherein the controller has first, second, and

third control inputs, the second control input coupled to the delay control input, and the controller includes: an oscillator having a frequency control input and an oscillator output, the frequency control input coupled to the first control input; a modulation signal generator having a generator input, in-phase differential generator outputs, and quadrature-phase differential generator outputs, the generator input coupled to the oscillator output, and the quadrature-phase differential generator outputs coupled to the differential delay inputs; and a first multiplexor circuit having a first multiplexor input, a second multiplexor input, a first multiplexor output, and a first selection input, the first multiplexor input coupled to a first one of the in-phase differential generator outputs, the second multiplexor input coupled to a first one of the differential delay outputs, the first multiplexor output coupled to the first one of the differential modulation control outputs, and the first selection input coupled to the third control input; and a second multiplexor circuit having a third multiplexor input, a fourth multiplexor input, a second multiplexor output, and a second selection input, the third multiplexor input coupled to a second one of the in-phase differential generator outputs, the fourth multiplexor input coupled to a second one of the differential delay outputs, the second multiplexor output coupled to the second one of the differential modulation control outputs, and the second selection input coupled to the third control input.

5. The apparatus of claim 1, further comprising a low pass filter coupled between the first differential terminals and the differential modulator circuit.

6. The apparatus of claim 1, wherein the controller has a control input, and the apparatus further comprises a clock recovery circuit coupled between the first differential terminals and the control input.

7. The apparatus of claim 1, wherein the differential modulator circuit is configurable to transmit the modulated first and second signals as full-duplex signals.

8. An apparatus comprising: a first controller having first differential modulation control outputs and including a programmable delay circuit, the first controller configurable to provide first differential modulation signals having a particular frequency at the first differential modulation control outputs; a first differential modulator circuit coupled between first differential terminals and second differential terminals, the first differential modulator circuit having first differential modulation control inputs coupled to the first differential modulation control outputs, and the first differential modulator circuit configurable to: modulate first differential signals at the first differential terminals with the first differential modulation signals having the particular frequency; provide the modulated first differential signals at the second differential terminals; modulate modulated second differential signals at the second differential terminals with the first differential modulation signals having the particular frequency to recover second differential signals; and provide the recovered second differential signals at the first differential terminals; a second controller having second differential modulation control outputs and configurable to provide second differential modulation signals having the particular frequency at the second differential modulation control outputs; a second differential modulator circuit coupled between third differential terminals and fourth differential terminals, the second differential modulator circuit having second differential modulation control inputs coupled to the second differential modulation control outputs, and the second differential modulator circuit configurable to: modulate the modulated first differential signals at the third differential terminals with the second differential modulation signals having the particular frequency to recover the first differential signals; provide the recovered first differential signals at the fourth differential terminals; modulate second differential signals at the fourth differential terminals with the second differential modulation signals having the particular frequency; and provide the modulated second differential signals at the third differential terminals.

9. The apparatus of claim 8, wherein the first programmable delay circuit is configurable to set a phase shift between the first and second differential modulation signals, in which the phase shift is based on a propagation delay between the second differential terminals and third differential

terminals.

10. The apparatus of claim 9, wherein the first controller has a control input, and the second controller includes a control output coupled to the control input, and the first controller configurable to: receive a control signal at the control input representing the second differential modulation signals; and determine the phase shift based on the control signal.

11. The apparatus of claim 9, wherein the first controller has a control input, and the apparatus further comprises a clock recovery circuit having inputs and a clock recovery output, the inputs of the clock recovery circuit coupled to the first terminals, the clock recovery output coupled to the control input, and the first controller configurable to: receive a control signal at the control input representing a phase relationship between the recovered second differential signals and the first differential modulation signals; and determine the phase shift based on the control signal.

12. The apparatus of claim 8, wherein: the first controller includes a first modulation signal generator having in-phase differential generator outputs coupled to the first differential modulation control outputs; and the second controller includes a second modulation signal generator having quadrature-phase differential generator outputs coupled to the second differential modulation control outputs.

13. The apparatus of claim 8, wherein the first differential modulator circuit is configurable to transmit the modulated first differential signals and the recovered second differential signals as full-duplex signals; and wherein the second differential modulator circuit is configurable to transmit the modulated second differential signals and the recovered first differential signals as full-duplex signals.

14. The apparatus of claim 8, further comprising an isolation device coupled between the second and third differential terminals.

15. The apparatus of claim 14, wherein: the first controller and the first differential modulator circuit have a first ground terminal; the second controller and the second differential modulator circuit have a second ground terminal; and the isolation device is configurable to isolate the first ground terminal from the second ground terminal.

16. The apparatus of claim 14, wherein: the first controller and the first differential modulator circuit have a first power supply terminal; the second controller and the second differential modulator circuit have a second power supply terminal; and the isolation device is configurable to isolate the first power supply terminal from the second power supply terminal.

17. The apparatus of claim 14, wherein: the first controller and the first differential modulator circuit are on a first semiconductor die; the second controller and the second differential modulator circuit are on a second semiconductor die; and the first and second dies are included within a same package.

18. The apparatus of claim 14, wherein the isolation device comprises a transformer having a primary winding coupled between the second differential terminals and a secondary winding coupled between the third differential terminals.

19. The apparatus of claim 14, wherein the isolation device comprises: a first transformer having a first primary winding and a first secondary winding, the first primary winding coupled between the second differential terminals; and a second transformer having a second primary winding and a second secondary winding, the second secondary winding coupled between the third differential terminals; and wherein the first secondary winding is coupled to the second primary winding.

20. The apparatus of claim 19, wherein the first secondary winding is coupled to the second primary winding via a transmission line or a capacitor.

21. An apparatus comprising: a controller having a modulation control output and including a programmable delay circuit, the controller configurable to provide a modulation signal having a particular frequency at the modulation control output; and a modulator circuit coupled between a first terminal and a second terminal, the modulator circuit having a modulation control input coupled to the modulation control output, and the modulator circuit configurable to: modulate a

first signal at the first terminal with the modulation signal having the particular frequency; provide the modulated first signal at the second terminal; modulate a second signal at the second terminal with the modulation signal having the particular frequency; and provide the modulated second signal at the first terminal.
